

do30_transf_aggregation

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Dataset Summaries

Some `daltoolbox` transformations are not instance-level edits but dataset-level restructuring steps — this deck introduces `aggregation`, the one preprocessing technique in the collection that changes the *shape* of the dataset itself, not just the values inside it.

1 TECHNIQUE • AGGREGATION

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Why Aggregation Is Different

Every other transformation in the `transf` collection — sampling, cleaning, scaling, encoding, smoothing, balancing, feature selection, dimensionality reduction — preserves the row-level structure of the dataset: each output row still corresponds to one input row (or a filtered/reweighted subset of them).

- **Aggregation collapses rows.** The output has one row per group, not per original observation.
- **It is useful before plotting, reporting, and grouped comparison** — exactly the kind of "one number per category" summary that feeds a bar chart or a stakeholder

table.

- **It closes the preprocessing pipeline.** Sampling, cleaning, scaling, and encoding all prepare individual rows for modeling; aggregation prepares grouped rows for summarizing.

Grouped Summaries — aggregation

Summarizes a dataset by groups through named aggregation expressions. The interface follows the same "configure, then transform" pattern as every other `transf` technique, but the expressions themselves borrow directly from `dplyr`-style summary syntax: `name = expression`.

```
agg <- aggregation(  
  "Species",  
  mean_sepal = mean(Sepal.Length),  
  sd_sepal   = sd(Sepal.Length),  
  n          = n()  
)  
  
iris_agg <- transform(agg, datasets::iris)
```

- First argument: the grouping attribute (`"Species"`).
- Remaining arguments: named expressions evaluated within each group — any summary function works (`mean` , `sd` , `n()` , `median` , custom functions).
- Output: one row per group, one column per named expression.

What to Observe

- Aggregation is a *dataset-level* restructuring step, not a row-level edit — a conceptual category of its own inside the `transf` collection.
- The named-expression interface scales to any number of summary statistics without changing the call shape.
- Natural next step after aggregation: feed the summarized table directly into the `graphics` comparison charts (bar, grouped bar, lollipop) covered in the visualization decks.